10TH CLASS BIOLOGY — NEW SYLLABUS (PECTAA)

Chapter 1 — Human Digestive System

Solved Exercise | New Book (PECTAA)

Exercise (Solved)

A. Select the correct answers for the following questions.

1. The three portions of the small intestine, in the correct order, are:

(a) Duodenum, jejunum, ileum (b) Ileum, duodenum, jejunum

(c) Colon, caecum, rectum (d) Caecum, colon, rectum

Answer: (a) — The small intestine is a highly coiled tube of nearly 7 m. Food passes first into the duodenum (about 25 cm), then into the jejunum (about 2.5 m) and lastly into the ileum (about 4 m). Colon, caecum and rectum are parts of the large intestine, not of the small intestine.

2. The wavelike movement of muscles that pushes food through the digestive system is called:

(a) Chemical digestion (b) Mechanical digestion

(c) Peristalsis (d) Absorption

Answer: (c) — When food enters the oesophagus, successive waves of contraction are produced in its muscular walls, and these waves push the food towards the stomach. The same wavelike movement carries food through the rest of the alimentary canal; it is called peristalsis.

3. The part of the digestive system where no chemical digestion takes place is:

(a) Oral cavity (b) Oesophagus

(c) Stomach (d) Duodenum

Answer: (b) — Only peristalsis takes place in the oesophagus, so no enzyme acts on food there. Chemical digestion starts in the oral cavity (salivary amylase), continues in the stomach (pepsin) and is completed in the duodenum, where the juices of the liver and pancreas act on food.

4. Which group of enzymes breaks up starches?

(a) Proteases (b) Lipases

(c) Amylases (d) Pepsin

Answer: (c) — Amylases break down starch into maltose. Proteases (such as pepsin and trypsin) digest proteins, and lipases digest fats.

StudySapiens.com

5. The pancreas produces digestive enzymes and releases them into:

(a) Colon (b) Gall bladder
(c) Liver (d) Duodenum

Answer: (d) — The secretion of the pancreas, called pancreatic juice, enters the duodenum through the pancreatic duct. This duct joins the common bile duct before it opens into the duodenum.

6. In stomach, pepsinogen is converted into pepsin by the action of:

(a) Bile salts (b) Hormones
(c) HCl (d) Bicarbonate

Answer: (c) — Gastric juice contains hydrochloric acid and the inactive enzyme pepsinogen. Hydrochloric acid converts this inactive pepsinogen into the active enzyme pepsin, which then breaks large proteins into shorter chains of amino acids called peptides.

7. In which part are the carbohydrates, lipids and proteins digested?

(a) Oral cavity (b) Stomach
(c) Small intestine (d) Large intestine

Answer: (c) — All three types of food are digested completely in the small intestine: pancreatic amylase digests starch, trypsin and chymotrypsin digest proteins, and lipase digests fats. The glands in the wall of the small intestine also secrete enzymes (maltase, sucrase, lactase and peptidases) for the complete digestion of food.

8. The enzyme trypsin is found in:

(a) Pancreatic juice (b) Bile
(c) Gastric juice (d) Saliva

Answer: (a) — Pancreatic juice contains trypsin, pancreatic amylase and lipase, which digest proteins, carbohydrates and lipids respectively. Bile contains no enzymes at all, gastric juice contains pepsin and saliva contains salivary amylase.

9. The common bile duct carries bile from gall bladder to:

(a) Stomach (b) Duodenum
(c) Colon (d) Pancreas

Answer: (b) — The common bile duct carries bile from the gall bladder into the duodenum, which is the first part of the small intestine. In the duodenum the bile salts break the large fat droplets into small droplets, and the pancreatic duct also opens here.

studysapiens.com

10. Why the small intestine has many folds and contains villi and microvilli?

(a) To slow down the passage of food (b) To promote food absorption

(c) To prevent the backflow of chyme (d) To produce enzymes

Answer: (b) — The folds and the millions of finger-like villi and microvilli on them provide a very large surface area for the absorption of digested food. Because of this large surface, the food molecules are quickly absorbed by diffusion and active transport.

B. Write short answers.

1. State the role of salivary glands in digestion.

Three pairs of salivary glands are attached to the oral cavity. They secrete saliva, which is a mixture of water, mucus and a digestive enzyme called salivary amylase. The water and mucus moisten the food pieces and make them soft, and salivary amylase breaks down the starch present in the food into maltose. This is the partial chemical digestion of carbohydrates, which begins in the oral cavity.

2. What do the terms swallowing and peristalsis mean?

Swallowing: after the physical and partial chemical digestion in the oral cavity, the food mass becomes a soft ball called bolus. The tongue pushes this bolus to the back of the oral cavity and it is swallowed into the pharynx. During swallowing the palate moves upward to close the opening of the nasal cavity, and the epiglottis closes the glottis, so the food does not enter the windpipe.

Peristalsis: it is the series of successive waves of muscular contraction in the walls of the alimentary canal. These waves push the food forward, from the oesophagus into the stomach and then through the intestines. It is peristalsis that moves food along the digestive system.

3. What are the functions of HCl in gastric juice?

Hydrochloric acid is present in the gastric juice of the stomach. Its functions are:

(i) It converts the inactive enzyme pepsinogen into the active enzyme pepsin, which then breaks large proteins into peptides.

(ii) It kills the pathogenic bacteria present in food, so it protects us from infection.

(iii) It provides an acidic medium which is necessary for the action of pepsin, and mucus present in the gastric juice neutralizes this acid near the stomach wall so that the wall is not digested.

4. State the role of pancreatic juice in digestion of food.

Pancreatic juice is secreted by the pancreas and reaches the duodenum through the pancreatic duct. It contains many enzymes, for example trypsin, pancreatic amylase and lipase, which digest proteins, carbohydrates and lipids respectively. It also contains sodium bicarbonate, which neutralizes the acidity of the chyme that comes from the stomach, so that the enzymes of the small intestine can work properly.

Enzyme of pancreatic juice	Substrate	Product
Pancreatic amylase	Starch	Maltose
Trypsin	Proteins	Peptides
Lipase	Fats (lipids)	Fatty acids and glycerol

5. Write a brief note on the structure of villi.

The inner wall of the small intestine is highly folded, and these folds carry millions of finger-like projections called villi (singular: villus). These folds and villi, along with the microvilli present on them, give the inner wall a very large surface area for the absorption of food. The wall of a villus is made of only a single layer of cells, so digested food can pass through it easily. Inside every villus there are blood capillaries and a lymph vessel called lacteal: glycerol and fatty acids enter the lacteals, while amino acids and simple sugars enter the blood capillaries.

6. What are the causes and symptoms of ulcers?

Peptic ulcers are sores which develop in the walls of the alimentary canal. The ulcer which develops in the wall of the stomach is called gastric ulcer, and it can also develop in the duodenum (duodenal ulcer) or in the oesophagus (oesophageal ulcer).

Symptoms: burning stomach pain, rush of saliva after regurgitation, loss of appetite, weight loss, and blood vomiting in severe cases.

Causes: infection due to the bacterium Helicobacter pylori, long-term use of nonsteroidal anti-inflammatory drugs (NSAIDs) such as painkillers, excessive use of alcohol, smoking and stress. Stress is not a direct cause, but it makes the symptoms worse.

C. Write answers in detail.

1. Describe the absorption of food in small intestine.

After the complete digestion of food, the end products such as amino acids, simple sugars, glycerol and fatty acids move from the alimentary canal into the circulatory system. This movement of digested food into the blood or lymph is called absorption, and most of it takes place in the small intestine.

The inner wall of the small intestine is highly folded, and these folds have millions of finger-like projections called villi. The folds, the villi and the microvilli present on them provide a very large surface area for absorption. The wall of a single villus is only one cell thick, and food molecules are absorbed through this surface by diffusion and active transport.

Route of fatty acids and glycerol: the glycerol and fatty acids present in the small intestine enter the lacteals (lymph vessels) of the villi. The lacteals carry them to the main lymph vessels, which empty these nutrients into the blood vessels near the heart.

Route of sugars and amino acids: amino acids and simple sugars enter the blood capillaries of the villi. These capillaries join to form the hepatic portal vein, which

carries them to the liver. From the liver, the hepatic vein carries these nutrients to the heart, and then they are supplied to all parts of the body. Water, mineral salts and vitamins are also absorbed here, and the material left undigested passes to the large intestine, where more water is absorbed from it.

2. Explain the role of oral cavity in the digestive system.

The space behind the mouth is called oral cavity or mouth cavity. It contains the teeth and the tongue, and the tongue has taste buds on its surface for the taste of food. The digestion of food begins here, in two ways.

Mechanical digestion: the teeth cut and grind the food into smaller pieces, and the tongue keeps the food between the teeth during this grinding. The food is then mixed with saliva and becomes a soft ball called bolus.

Chemical digestion: three pairs of salivary glands are attached to the oral cavity and they secrete saliva, which is a mixture of water, mucus and salivary amylase. Water and mucus moisten the food pieces, while salivary amylase breaks the starch present in the food into maltose. In this way the partial chemical digestion of carbohydrates takes place in the oral cavity.

After this, the tongue pushes the bolus to the back of the oral cavity, and the food is swallowed into the pharynx. The palate moves up to close the nasal opening, so food does not go into the nose. It should be noted that no absorption of food takes place in the oral cavity; only grinding, moistening and partial digestion of starch take place here.

3. Explain the structure and functions of stomach.

Structure: the stomach is a J-shaped organ located in the upper left side of the abdominal cavity, below the diaphragm. The part just after the oesophagus is called the cardiac end, and the part just before the small intestine is called the pyloric end. At the junction of the oesophagus and the stomach there is a ring of muscles called the cardiac sphincter, which prevents the food from flowing back into the oesophagus. Similarly, the pyloric sphincter is present between the stomach and the small intestine. The walls of the stomach are made of thick smooth muscles, and many small gastric glands are present in its inner walls.

Functions:

(i) Storage of food: food usually remains in the stomach for three to four hours.

(ii) Mechanical digestion: when food arrives, the smooth muscles contract strongly and churn the food into smaller particles. Heat produced during churning also helps to melt the lipids.

(iii) Chemical digestion of proteins: the gastric glands secrete gastric juice, which contains hydrochloric acid, pepsinogen and mucus. Hydrochloric acid converts pepsinogen into active pepsin, and pepsin breaks large proteins into peptides. Hydrochloric acid also kills pathogenic bacteria present in food, while mucus forms a protective layer and neutralizes the acid near the wall so that the stomach is not digested by its own enzymes.

(iv) Formation of chyme: due to all these actions the food becomes a soup-like mixture called chyme. The pyloric sphincter controls its flow, and each time it opens

about 5 to 15 mL of chyme passes into the small intestine.

4. Justify the importance of liver and pancreas in digestion.

Liver: the liver is a large organ located to the right of the stomach and it weighs about 1.5 kg in an adult. It secretes bile, which is stored in the gall bladder and carried by the common bile duct into the duodenum. Bile contains no enzymes, but its salts break large fats into small droplets and keep these droplets separate from one another. This process is called emulsification. In this way the surface area of fats is increased many times, so lipase can attack the fats easily and digest them quickly. Without bile, fats would remain in large drops and their digestion would be very difficult.

The liver also stores glucose as glycogen, stores fat-soluble vitamins, breaks amino acids (forming ammonia) and converts ammonia into less toxic urea, breaks toxic substances such as alcohol, breaks old red blood cells, prepares vitamin A from carotene and produces heat in cold weather.

Pancreas: the pancreas sends its secretion, called pancreatic juice, into the duodenum through the pancreatic duct. This juice contains trypsin and chymotrypsin for proteins, pancreatic amylase for starch and lipase for fats, so it digests all three main types of food. It also contains sodium bicarbonate, which neutralizes the acidic chyme coming from the stomach, so the enzymes of the intestine can work in the correct medium.

Enzyme	Site of action	Substrate	Product
Salivary amylase	Oral cavity	Starch	Maltose
Pepsin	Stomach	Proteins	Peptides
Trypsin, chymotrypsin	Small intestine	Proteins	Peptides
Pancreatic amylase	Small intestine	Starch	Maltose
Lipase	Small intestine	Fats	Fatty acids and glycerol
Peptidases, maltase, sucrase, lactase	Small intestine	Peptides, maltose, sucrose, lactose	Amino acids, glucose, glucose and fructose, glucose and galactose

D. Inquisitive Questions

1. Why does someone with liver disease have difficulty in digesting fats?

The liver secretes bile, and bile salts break the large fat droplets of food into very small droplets (emulsification). Only after this emulsification can the enzyme lipase attack the fats and digest them. In a person with liver disease the production or the flow of bile is reduced, so the fats of the food are not emulsified properly. Consequently lipase cannot digest them efficiently, and the person feels heaviness in the stomach, indigestion and oily stools after eating fatty food.

2. Why is it important to maintain a healthy balance of gut bacteria?

A very large number of useful bacteria live in our large intestine, and they help us in many ways. They break down some of the undigested food and make it easier to pass out, they prepare some vitamins such as vitamin K, and they protect us from harmful germs by competing with them for food and space. They also keep the lining of the gut and the immune system healthy. If this balance is disturbed, for example by taking antibiotics again and again, the harmful bacteria increase and cause problems such as diarrhoea, constipation, gas and infections. That is why we should take fibre-rich food, clean water and fresh curd, and avoid unnecessary medicines, so that the useful bacteria stay in balance.

3. How can a malfunction in the cardiac sphincter lead to acid reflux?

The cardiac sphincter is a ring of muscles at the junction of the oesophagus and the stomach. Normally it stays closed and prevents the food and the acidic chyme of the stomach from flowing back into the oesophagus. If this sphincter becomes weak or relaxes at the wrong time, the acidic contents of the stomach move back into the oesophagus, which is called acid reflux. The oesophagus has no protective mucus lining like the stomach, so the acid causes a burning sensation in the chest, called heartburn, and a sour taste in the mouth. That is why the person feels pain after a heavy meal or on lying down.

4. Why is it necessary to store bile in the gallbladder before it is released?

The liver secretes bile continuously, but it is needed only when fatty food reaches the duodenum, that is, only two or three times a day. The gall bladder stores this bile between the meals and also concentrates it by absorbing water from it. When fatty food enters the duodenum, the gall bladder contracts and sends a larger amount of concentrated bile into the duodenum through the common bile duct. In this way enough bile is available at the right time for the emulsification and digestion of fats. If the bile were released all the time, it would be wasted and the fats would still not be digested properly.

studysapiens.com